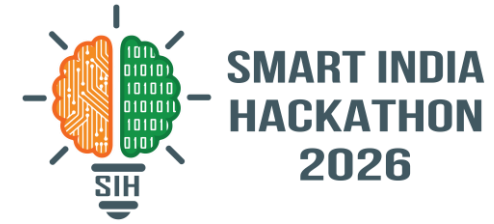
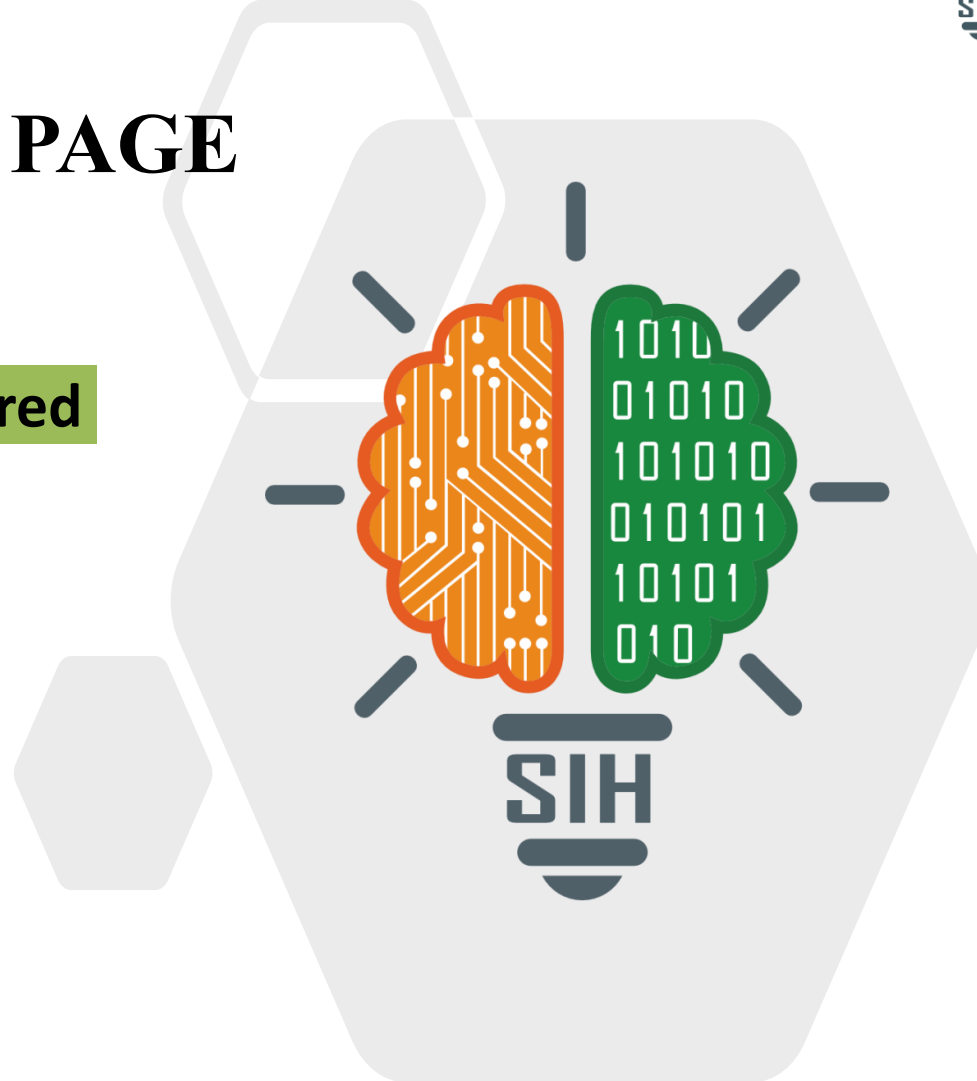


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID- **SIH26189**
- Problem Statement Title- **AI-Powered Criminal Network Analysis System**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID-
- Team Name- **Gits N' Roses**



IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

➤ **The Problem:**

Criminal networks today are organized and interconnected — spanning associates, financial channels, communication links, and locations. Law enforcement has the data to catch them, but it's scattered across seven disconnected sources:

- FIRs & police reports
- Call Detail Records (CDRs)
- Financial transaction records
- Surveillance reports
- Social media intelligence
- Criminal history databases
- Intelligence agency reports

The result: fragmented, unstructured data across multiple systems → slow, manual, error-prone analysis → critical connections get missed.

➤ **Detailed explanation of our solution:**

- Collect and process data from multiple sources
- Extract important entities: people, locations, vehicles, phone numbers, organizations
- Build relationship maps showing how entities are connected
- Identify key individuals who play influential roles in the network
- Detect suspicious patterns & unusual activities & stores it using blockchain
- Provide visual and analytical insights to investigators

➤ **How it addresses the problem:**

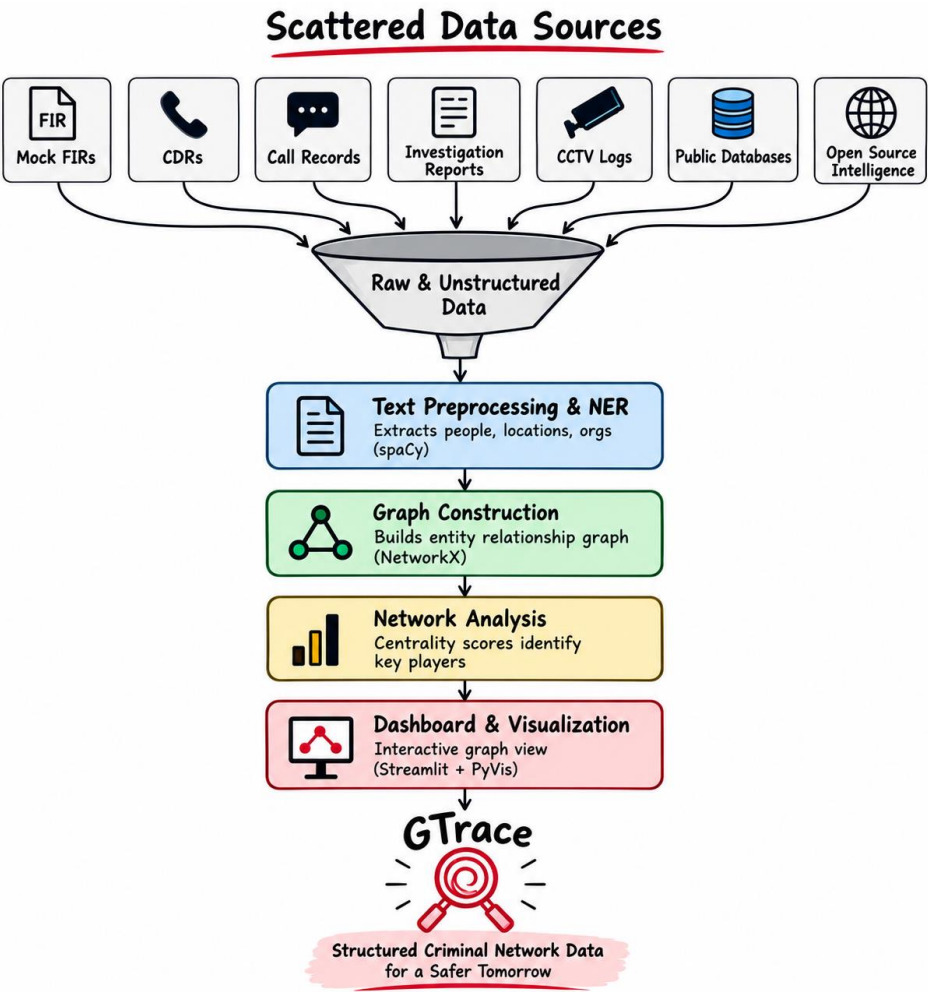
- Investigators currently rely on manual cross-referencing of thousands of documents, which is slow and misses connections.
- Our system automates entity extraction and relationship discovery, surfacing hidden links and ranking individuals by network importance — cutting investigation time and reducing missed connections.

- **Innovation/uniqueness:** Combines pretrained NLP (spaCy) for entity extraction with graph theory (NetworkX centrality algorithms) for influence scoring, delivered through an interactive visual dashboard rather than requiring investigators to read raw reports.

❑ Technologies to be used:

Tool	Purpose
Python	Core language connecting the pipeline
spaCy	Named entity extraction (NER)
Regex	Catches phone numbers, plates
NetworkX	Graph building + centrality analysis
PyVis	Interactive graph visualization
hashlib (Python built-in)	Blockchain-inspired tamper-evident audit log
Streamlit	Dashboard/UI

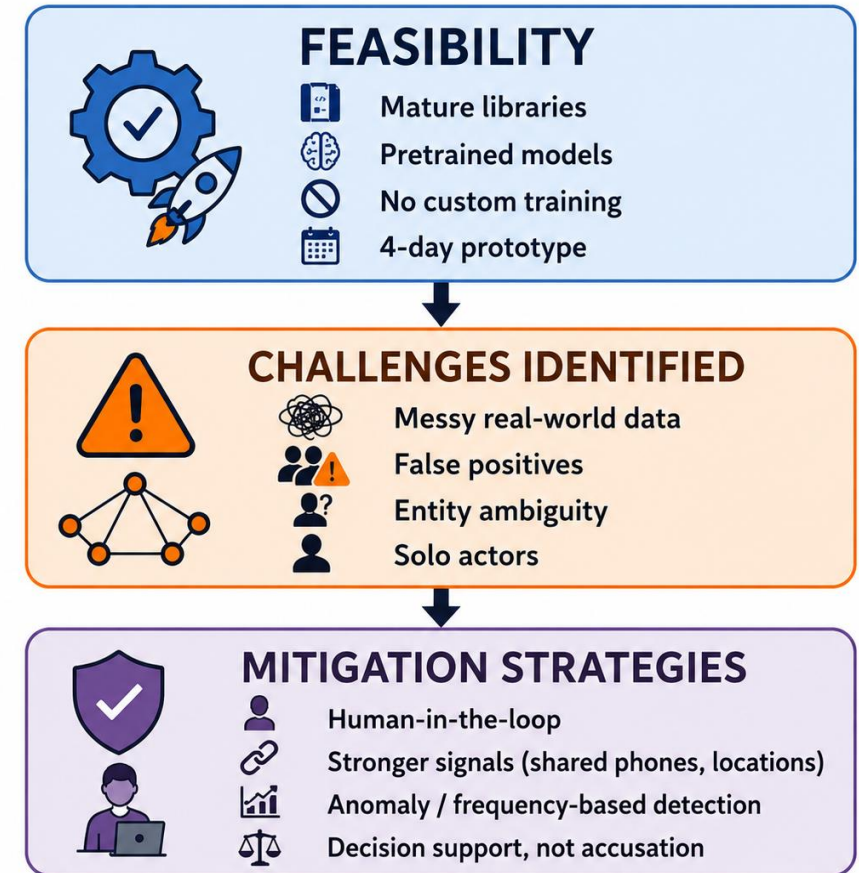
❑ Process flow:



- **Feasibility:** High for a working prototype — every pipeline stage uses a mature, pretrained, or well-documented library (no custom model training required), making it buildable within a short timeline.

- **Challenges identified:** Real-world crime data is far messier than clean mock data (typos, regional language, inconsistent formats) Co-occurrence-based connections can produce false positives (two people mentioned together without an actual criminal link) Entity disambiguation (same name referring to different people) isn't handled by default NER Solo/independent criminals with no shared connections are harder to flag through pure network analysis

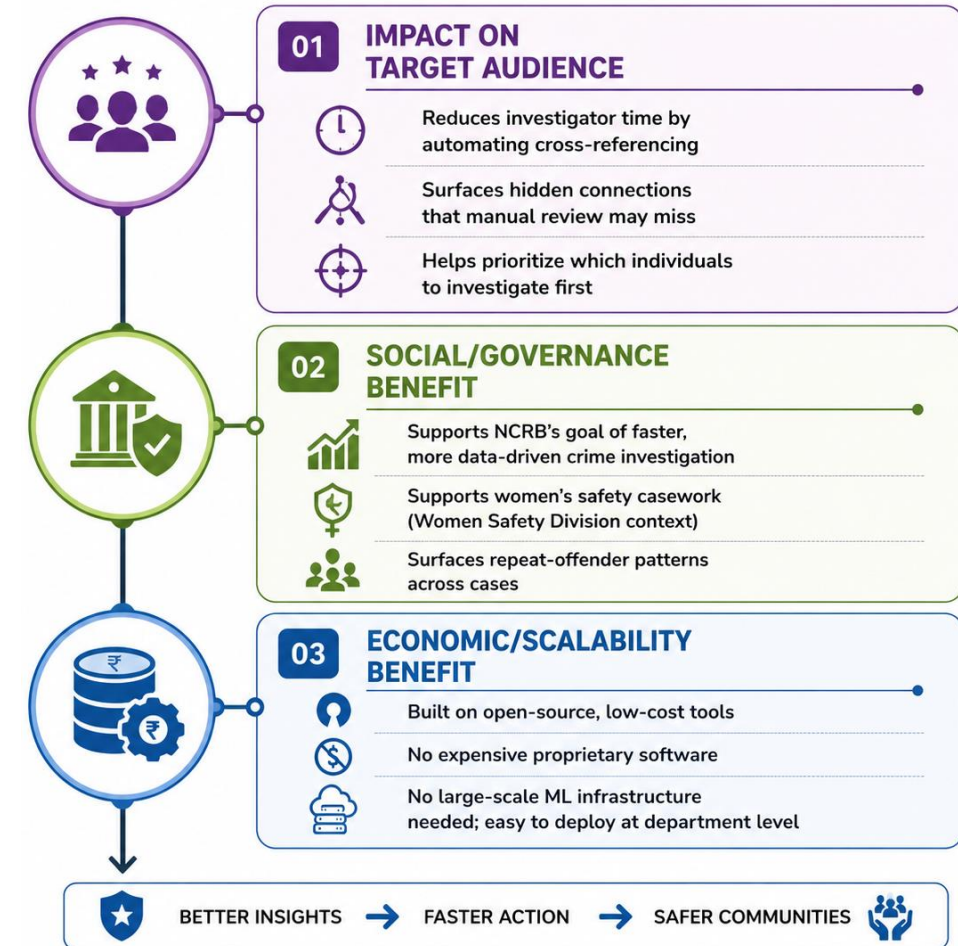
Mitigation strategies: Position the system as investigator decision-support, not an automated accusation tool — final judgment stays human Combine co-occurrence with stronger signals (shared phone numbers, locations) to reduce false positives Extend the model with anomaly/frequency-based detection to also catch solo actors, not just networked ones Plan a human-in-the-loop verification step before any flagged connection is acted on



❑ **Impact on target audience:** Reduces investigator time spent manually cross-referencing records; surfaces hidden connections that manual review would likely miss; helps prioritize which individuals to investigate first.

❑ **Social/governance benefit:** Supports NCRB's broader goal of faster, more data-driven crime investigation; can be extended to support women's safety casework (given the Women Safety Division context) by surfacing repeat-offender patterns across cases.

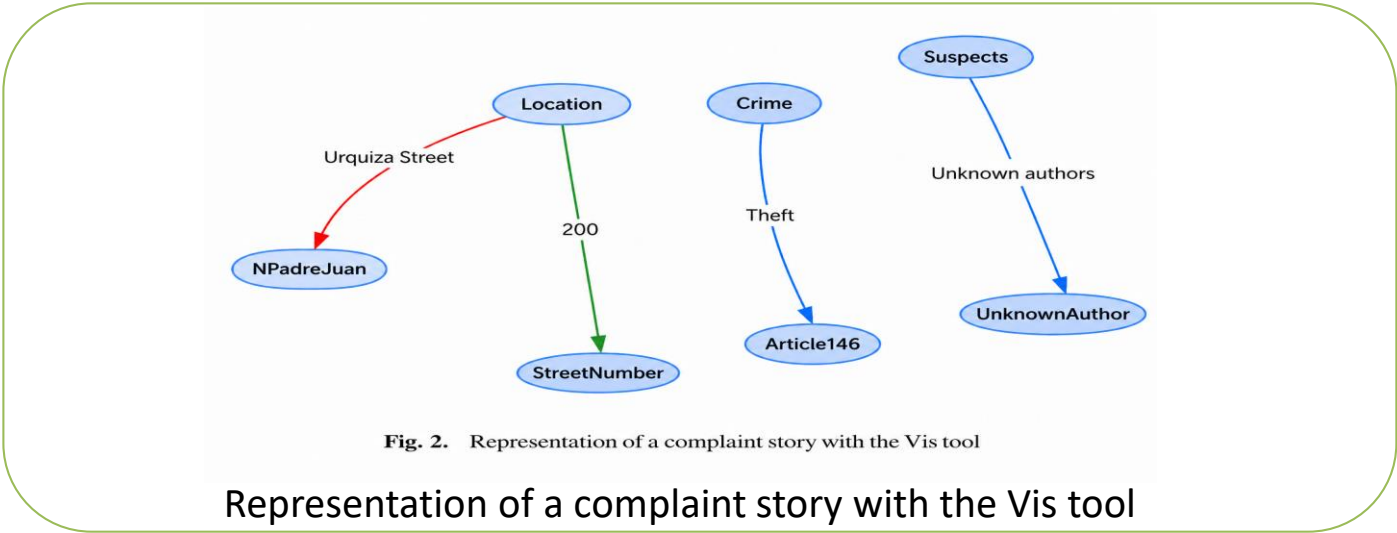
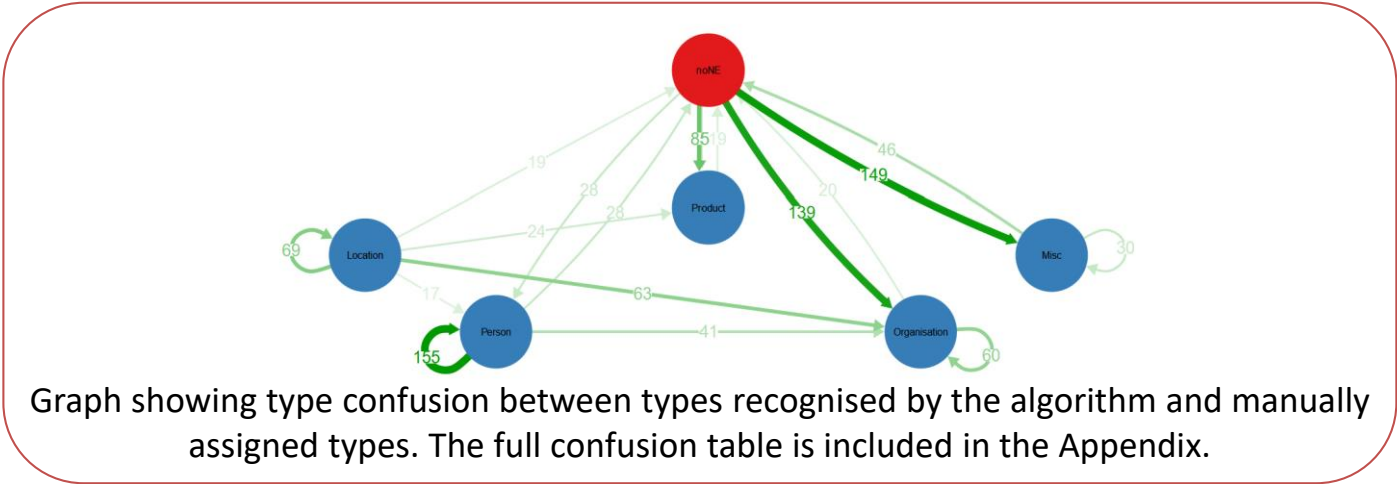
❑ **Economic/scalability benefit:** Built on open-source, low-cost tools — no expensive proprietary software or large-scale ML infrastructure needed to deploy at a department level.



➤ Schraagen et al. (2017) confirm NER is effective on real police complaint text but identify systematic recall/confusion errors — a limitation our system acknowledges and would need to address with domain-specific fine-tuning in production.

<https://dspace.library.uu.nl/bitstream/handle/1874/356185/01.NERclinjournal.pdf?sequence=1>

➤ Álvarez & Antonelli (2024) apply spaCy-based NER to police complaints to automatically generate link maps between related cases — validating the core NER-to-graph approach our system extends with network-wide centrality analysis.



<https://digital.cic.gba.gob.ar/server/api/core/bitstreams/6bdf5e45-2f17-4db6-af51-062abafe4743/content>